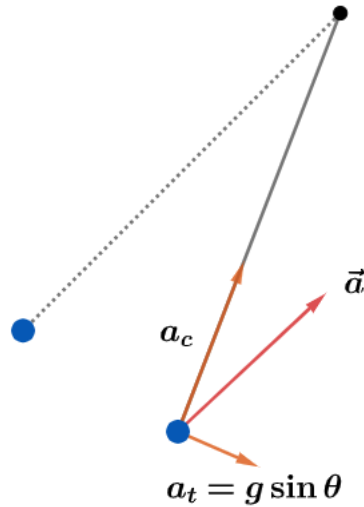


2023 F=ma Exam: Problem 2

Kevin S. Huang



The pendulum has a nonzero speed, so it has a centripetal acceleration $\vec{a}_c$ directed towards the center of rotation. The pendulum also has a tangential acceleration $\vec{a}_t$ due to gravity. The total acceleration $\vec{a}$ is given by the vector sum:

$$\vec{a} = \vec{a}_c + \vec{a}_t$$

so the answer is B.